Development and Analysis of a Thermodynamic Table Program

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Literature Review

This report presents the development of a computer program designed to display thermodynamic tables and determine the thermodynamic state of a substance based on user-defined temperature and pressure inputs. The program, implemented using the C++ programming language, utilizes arrays to store thermodynamic properties, including temperature, pressure, enthalpy of liquid, and enthalpy of vaporization.

The program provides a user-friendly interface that allows users to display thermodynamic tables and compute enthalpy values. Based on the input conditions, the program automatically identifies the state of the substance as saturated, compressed, or superheated. In saturated conditions, the vapor quality factor is employed to accurately calculate thermodynamic properties. This automated approach significantly reduces manual effort, improves computational accuracy, and enhances learning and practical understanding of thermodynamic concepts.

1. Introduction

Thermodynamics is a fundamental branch of physics and engineering that deals with the relationships between heat, work, and energy [1]. It provides essential principles for analyzing and designing energy systems, including engines, refrigeration cycles, power plants, and chemical processes.

The foundation of thermodynamics is built upon four fundamental laws:

1. **Zeroth Law of Thermodynamics:** Establishes the concept of temperature and thermal equilibrium.
2. **First Law of Thermodynamics:** States the conservation of energy, indicating that energy cannot be created or destroyed, only transformed from one form to another.
3. **Second Law of Thermodynamics:** Describes the increase of entropy and the direction of natural processes.
4. **Third Law of Thermodynamics:** Defines the behavior of entropy at absolute zero temperature.

These principles form the theoretical framework necessary for understanding energy interactions in natural and engineered systems [1].

2. Thermodynamic Tables

Thermodynamic tables are essential references in mechanical engineering and thermodynamic analysis [1,3]. They provide structured data for key thermodynamic properties, such as temperature, pressure, specific volume, enthalpy, and entropy. These tables enable engineers and students to evaluate system performance and analyze thermal processes efficiently.

By utilizing thermodynamic tables, engineers can perform precise calculations, optimize system performance, and design efficient thermal systems. These tables effectively bridge the gap between theoretical models and practical engineering applications.

3. Water as an Example of a Thermodynamic Table

Water is widely used as a reference substance in thermodynamic studies due to its well-documented properties and wide range of engineering applications [1,3]. Important features that make water particularly suitable include:

1. Phase transitions between solid, liquid, and vapor states.
2. Well-defined critical point (374°C, 22.1 MPa).
3. Well-defined triple point (0.01°C, 611.657 Pa).
4. Wide operational temperature range.
5. Familiarity and availability.

These characteristics make water an ideal model for understanding thermodynamic behavior and phase change phenomena.

4. States of Water

In thermodynamics, water can exist in several states depending on temperature and pressure:

- **Saturated Water:** Exists at the boiling point corresponding to a given pressure.
- **Superheated Vapor:** Vapor heated above its saturation temperature at a given pressure.
- **Compressed Liquid:** Liquid subjected to pressure higher than its saturation pressure.

Understanding these states is crucial for analyzing steam power cycles, refrigeration systems, and heat transfer processes.

5. Thermodynamic Properties

Several thermodynamic properties are essential for system analysis:

- **Quality (x):** Represents the mass fraction of vapor in a saturated liquid-vapor mixture.
- **Enthalpy (h):** Represents the total energy content of a system.
- **Specific Volume (v):** Volume per unit mass.
- **Entropy (s):** Measures disorder and energy dispersal within a system.

For saturated mixtures, enthalpy is calculated using:

$h = h_f + x \cdot h_{fg}$ where hf is the enthalpy of saturated liquid and hfg is the enthalpy of vaporization.

6. Problem Statement

Manual use of thermodynamic tables is time-consuming and prone to error due to the large volume of data, interpolation requirements, and complexity of state determination [1]. These challenges hinder efficient thermodynamic analysis and system optimization.

7. Proposed Solution

To overcome these challenges, a C++ program was developed to automate thermodynamic property calculations [2]. The program allows users to input temperature and pressure values, determines the thermodynamic state, and computes the required properties using stored data arrays and interpolation methods.

This automated approach improves efficiency, accuracy, and usability, making it a valuable educational and engineering tool.

8.Output Cases and Discussion

The developed C++ program provides different outputs depending on the entered temperature and pressure values. These outputs are categorized into four main cases, which demonstrate the program's ability to correctly determine the thermodynamic state of water and calculate the corresponding properties. The following cases are illustrated in Figures 1–4.

Case 1: Saturated State

When the input temperature and pressure correspond to values available directly in the thermodynamic table, the program identifies the state as saturated. In this case, the user is asked to enter the vapor quality factor *(x)*, and the program calculates the thermodynamic properties such as enthalpy, specific volume, and entropy using the relation:

$$y = y_f + x \cdot y_{fg}$$

Case 1:

```
65.00        25.0430
70.00        31.2020
75.00        38.5970
80.00        47.4160
```

```
Temp and pressure are same
```

```
Enter 2 to find values of water: 2
Enter the value of T and P: 70 31.2020
It is Saturated, so enter the value of x (must be 0-1): 0.2
values for x = 0.20 is:
 enthalpy = 759.6700 kj/kg
 specific volume =1.0076 m^3/kg
 entropy = 2.3149 kj/kg*kmol
 Do you want to continue calculating values of water? (1 for yes, 0 for no):
```

Figure 1 Saturated State Output

The output clearly displays the calculated properties, as shown in Figure 1, confirming the correctness of the implemented equations and data handling.

Case 2: Compressed Liquid State

If the entered pressure is higher than the saturation pressure at a given temperature, the program identifies the state as compressed liquid. In this situation, the exact values are not directly available in the table, and the program informs the user that interpolation is required for more accurate results. This behavior is demonstrated in Figure 2, where the program correctly classifies the state and provides appropriate guidance.

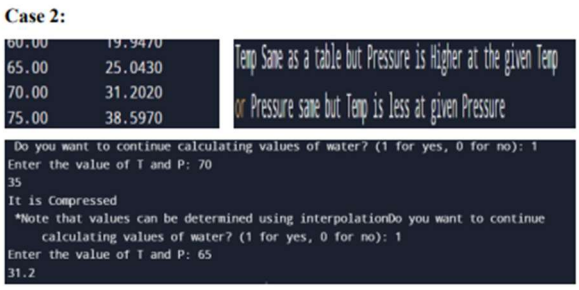

Figure 2 Saturated State Output

Case 3: Superheated Vapor State

When the entered temperature is higher than the saturation temperature corresponding to the given pressure, the program classifies the state as superheated vapor. Since these values are outside the saturated region, interpolation or superheated tables are needed to determine precise thermodynamic properties. As shown in Figure 3, the program successfully identifies this state and displays a suitable message to the user.

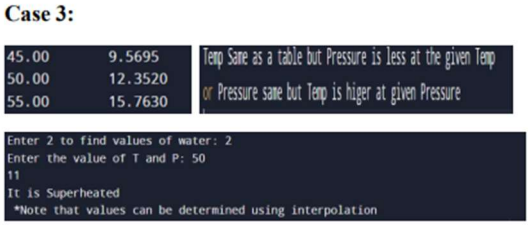

Figure 3 Superheated Vapor Output

Case 4: Out-of-Range Values

If the entered temperature and pressure values fall outside the available data range of the thermodynamic table, the program outputs a message indicating that the values are not found in the table. This case, illustrated in Figure 4, highlights the robustness of the program in handling invalid or extreme inputs and preventing incorrect calculations.

Overall, these four cases demonstrate the reliability, accuracy, and flexibility of the developed program in analyzing different thermodynamic conditions and guiding the user toward correct interpretations of the results.

The developed program successfully demonstrates accurate determination of thermodynamic states and properties [2,3]. It simplifies data retrieval, reduces manual computation time, and minimizes errors. The program also enhances conceptual understanding of thermodynamic principles and supports engineering problem-solving tasks.

Case 4:

```
The values are higher than what is in the table or both are not
    in the table
```

```
95.00      84.6090
100.00     101.4200
```

```
Enter 2 to find values of water: 2
Enter the value of T and P: 102
105
Values not found in the table.
```

```
Enter 2 to find values of water: 2
Enter the value of T and P: 72
40
Values not found in the table.
```

Figure 4 Out-of-Range Values Output

9. Conclusion

This project highlights the importance of thermodynamic tables in engineering analysis and demonstrates the effectiveness of computational tools in simplifying complex calculations [1]. The developed C++ program enhances learning outcomes, improves computational efficiency, and provides a reliable platform for thermodynamic analysis [2,3].

References

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